Mid-Term Evaluation Report

Deep Learning Detection of Fast radio burst signals in Radio Telescopes

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1. Paper Review

(i) The Second CHIME/FRB Catalog of Fast Radio Bursts (2026) - CHIME/FRB Collaboration

- The dataset consists of a comprehensive catalog of **4,539 Fast Radio Bursts (FRBs)** detected between 25 July 2018 and 15 September 2023, originating from **3,641 unique sources,** including **981 bursts from 83 repeating FRB sources**.
- For each event, the catalog provides total-intensity dynamic spectra covering the **400–800 MHz** frequency range with a time resolution of **0.983 ms,** along with fitted burst morphology parameters such as arrival time, dispersion measure (DM), pulse width, scattering timescale, and flux density.
- The catalog includes all bursts from the previous release, reprocessed using a uniform and improved analysis pipeline, enabling consistent characterization and comparison across the entire FRB population.

(ii) Connor & van Leeuwen (2018)

- Real-time FRB/RFI classification for single-pulse candidates using a tree-like **multi-branch DNN** (frequency-time image, DM-time array, pulse profile, multi-beam info).
- Trained model on real RFI triggers plus simulated FRBs injected in real noise. Recall/accuracy exceeds 99% above ~8σ S/N, ~100μs/candidate. Closest prior-art analogue to this project's single frequency-time-image CNN.

(iii) Devansh, Kshitij, Sarah, Duncan & Nathaniel - FETCH (2020)

- Developed a **telescope-agnostic classifier** using **transfer learning** from ImageNet-pretrained CNNs (VGG16/19, DenseNet), fusing frequency-time and DM-time images (256×256).
- They used **Multiplicative fusion** which generalized far better than concatenation (99.8% vs. 77.6% val. accuracy). Trained 11 different models each having accuracy/recall/F-score bar greater than 99.5%.

(iv) Arni, Blanco & Prabhu (2025)

- They worked on a different, downstream problem: **repeater vs. non-repeater** classification of already-confirmed CHIME/FRB Catalog bursts via a Supervised VAE on **17 physical features** (16-D latent space).
- F2 score was 0.9807 while prior best was 0.8180, with interpretable latent-space separation. Relevant mainly as a possible later stage once FRB/RFI screening is validated, not to the current methodology.

2. Problem Statement & Motivation

Fast Radio Bursts (FRBs) are **extremely short radio signals** coming from distant galaxies, but their exact origin is still not fully understood. Studying FRBs requires collecting and analyzing large numbers of reliable burst detections. Modern radio telescopes such as CHIME, SKA, and Apertif produce huge numbers of candidate signals every day, but most of these are caused by radio-frequency interference (RFI) or other noise and artefacts.

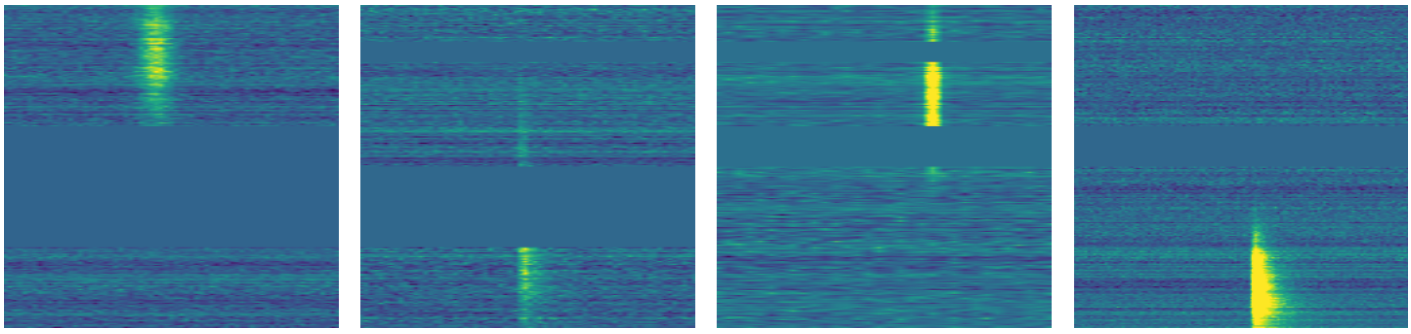
Manually checking such a large number of candidates is time-consuming and difficult to scale. Therefore, this project aims to develop an automated binary classification system that can distinguish between genuine FRBs and RFI/noise.

The system uses dedispersed dynamic spectra (frequency-vs-time images) as input to a Convolutional Neural Network (CNN), which can learn the characteristic visual patterns of FRBs directly from these images. The goal is to create a fast first-pass filter that can reduce the number of candidates requiring detailed analysis.

3. Approach / Methodology

3.1 FRB simulation

- Simulated **four** different sets of FRBs each run having som improvements over previous one. Thes first run was **baseline simulation.** Second on I've added **real data-texture and used fitburst module** to do simulation. In the third one I've added **jitters in position of burst** which earlier was at centre of every image. In the fourth one added some initials checks to rule out **duplicated entries** and also those having **NAN/Null** values so the catalogue is clean.
- The simulation uses CHIME/FRB's **fitburst** physics model. Key burst parameters such as DM, width, scattering, spectral index, and fluence are sampled from real events in the CHIME/FRB Catalog 2, making the simulated bursts more representative of observed FRBs.
- Each burst is generated on a **256-channel frequency grid** covering 400–800 MHz, with the time resolution adjusted based on the burst. The burst position is also randomized in time to avoid the CNN learning its position instead of its actual signal characteristics.

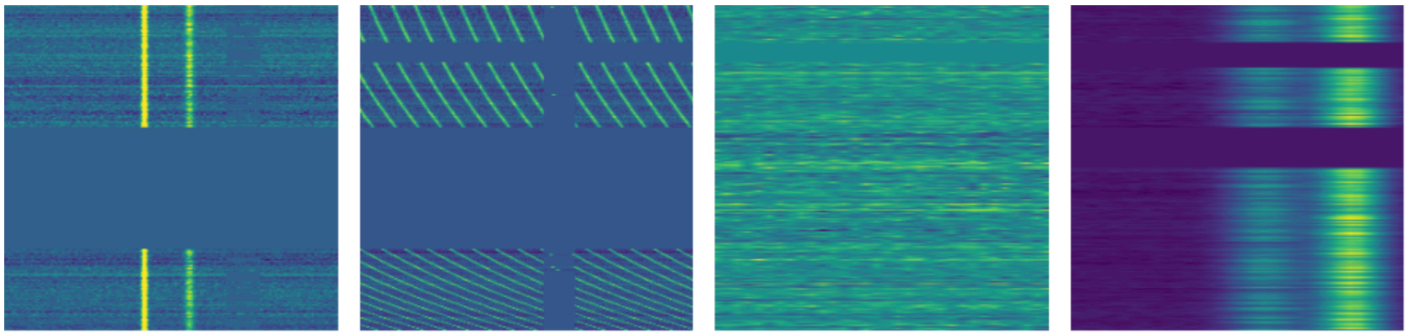


Samples of simulated FRBs

- To make the simulated data more realistic, the pipeline adds channel-dependent **bandpass effects, frequency ripple, radiometric noise, masked RFI bands etc.**

3.2 RFI simulation

- The same noise-generation code used for the FRB class is also used for the negative class. This helps prevent the CNN from simply learning differences between the two different data-generation pipelines. The window sizes are also sampled to match the FRB data distribution.
- The negative class includes pure noise and several realistic types of RFI, including **narrowband, aircraft, impulsive, periodic RFI, and complex RFI storms** that mimic a dispersion sweep. A real pulsar's periodic dispersed signal is also included as a challenging negative example. Additional effects such as scintillation, gain drift, time dropouts, and ADC quantization are added to further increase the diversity of the simulated data.



Samples og simulated RFIs

3.3 Dataset, preprocessing & split

- A total of **9,081** simulated images were generated, consisting of **4,081 FRB and 5,000 RFI** samples. All images passed the pre-training NaN/Inf quality check.
- The time axis of each dynamic spectrum was resized to 256 bins using linear interpolation. Each image was then normalized using its median and MAD, excluding masked-zero pixels, and the values were clipped to the range ±8.
- The data was divided using a **stratified 70/15/15 split** with seed 42, resulting in **6,357 training, 1,362 validation, and 1,362 test images.**
- The training data uses **WeightedRandomSampler** and **pos_weight** to handle class imbalance, while the validation and test sets retain their natural class distribution.

3.4 CNN, training & evaluation

- **PyTorch-based CNN** model, was developed with **four** convolutional blocks (16 → 32 → 64 → 128 filters), using **convolution, batch normalization, ReLU activation, and pooling.** An adaptive average pooling layer is followed by a fully connected head (128 → 128 → 1) with 0.3 dropout. The model takes 256×256 single-channel dynamic spectra as input.

- The model was trained using **Adam optimizer** with a **learning rate** of 1e-3 and **weight decay** of 1e-4, with a **batch size** of 32. Gradient clipping (5.0), learning-rate reduction on plateau, and **early stopping** based on validation F1- score (patience 6) were used. Training ran for **22 epochs**, with the best checkpoint obtained at **epoch 16.**

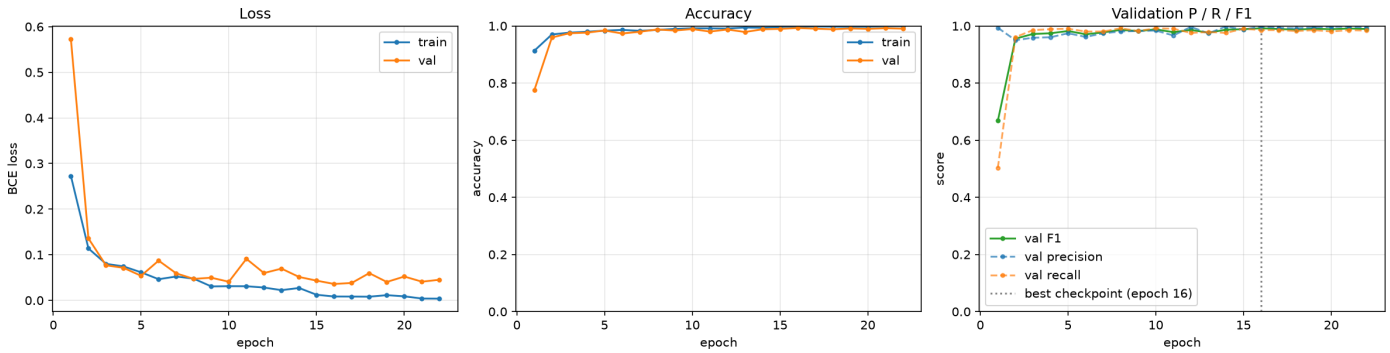
- The final model was evaluated once on the held-out test set using **accuracy, precision, recall, and F1-score.** A confusion matrix, ROC and precision-recall curves, classification report, and qualitative examples of true/false positives and negatives were also generated.

4. Results

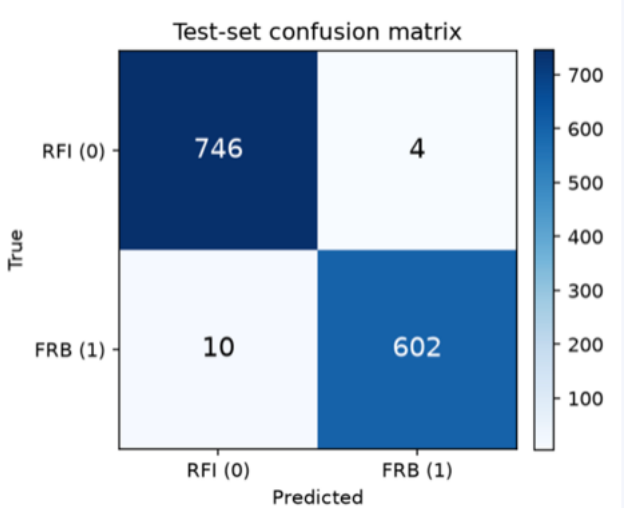
- Training converged smoothly where t**rain loss 0.273→0.004** (epochs 1→22), validation F1 peaked at **0.992** (epoch 16,_selected checkpoint). Validation loss stayed low (≈0.04–0.09) with mild noise hence **no sign of divergence or_major overfitting within this run.**

- The confusion_that out of_images only_**misclassified** matrix shows total 1362 test **14 were** by the model.

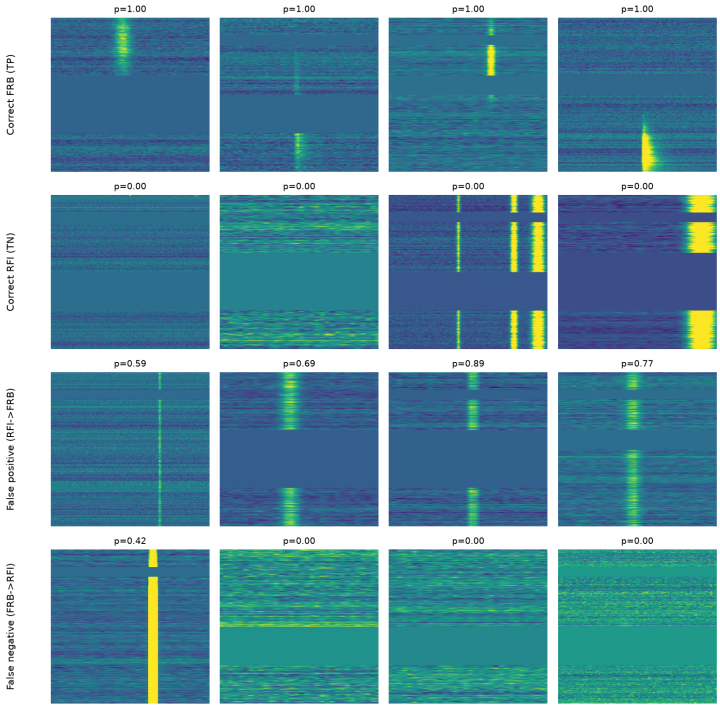
Metric	Value	Metric	Value
Accuracy	0.990	F1-score	0.989
Precision	0.993	Confusion matrix	TP 602, FP 4, FN 10, TN 746



Training curves; test-set confusion matrix; ROC (AUC 0.998) curve.



Confusion matrix of test set



example test predictions (p = model probability of FRB), correct FRB and RFI calls, plus false positives and false negatives.

Conclusion: true positives/negatives are classified with near-certain probability; false positives are narrowband/periodic RFI resembling a burst ridge; false negatives are dominated by low-S/N or borderline bursts, consistent with the RFI simulator's deliberately “hard negative” categories (Section 3.2).

Status of results

- **Completed:** end-to-end simulation of both classes, a full CNN training run with early stopping/ checkpoint selection, and a full test-set evaluation (metrics, ROC/PR, qualitative grid), all reproduced above from this project's own saved outputs.
- **Preliminary / single-run:** all metrics come from one training run, one fixed seed, and one simulated-dataset draw — no repeated runs or uncertainty estimate yet.
- **Limitation:** training and test data are entirely simulated (parameters grounded in real CHIME/FRB Catalog 2, but images are synthetic); no evaluation on real telescope candidates or RFI logs yet — the main gap vs. Connor & van Leeuwen and FETCH, both of which validate on real data.

5. Next Steps

- Repeat model training with techniques like Data augmentation, noise hardening, to increase both quality and quantity of data so as to reduce overfitting and make model more generalized
- Repeat the training with multiple random seeds and different dataset splits, then report the mean ± variation of the main evaluation metrics instead of relying on a single run.

- Obtain the original simulated FRB **.h5** files from **CANFAR**, either through the available API or by downloading the full dataset(~ 60 GB), to compare the model with more realistic data.
- Collect or generate more realistic RFI candidates and evaluate the current CNN on them to understand how well it generalizes beyond the simulated RFI used during training.
- Study and implement the FRB simulation approach used in *Frabjous: Deep Learning Fast Radio Burst Morphologies* and compare its simulated bursts with the current simulation pipeline.
- Experiment with pretrained CNN architectures, following approaches such as FETCH, and compare their performance with the current CNN trained from scratch.

6. References

1. Connor, L. & van Leeuwen, J. (2018). "*Applying Deep Learning to Fast Radio Burst Classification.*" (Section 1) arXiv:1803.03084.

2. Agarwal, D., Aggarwal, K., Burke-Spolaor, S., Lorimer, D. R., & Garver-Daniels, N. (2019/2020). "*FETCH: A deep-learning based classifier for fast transient classification.*" (Section 1) arXiv:1902.06343.

3. Arni, R., Blanco, C., & Prabhu, A. (2025). "*Using Deep Learning for Robust Classification of Fast Radio Bursts.*" Studied paper; sVAE-based repeater/non-repeater classification, relevant to downstream future work (Section 5) arXiv:2511.02634.

4. Ajay Kumar, Ashish A. Mahabal and Shriharsh P. Tendulkar "*Frabjous: Deep Learning Fast Radio Burst Morphologies*" (Section 5) arXiv:2507.14854.

5. CHIME/FRB Collaboration et al. (2021). The First CHIME/FRB Fast Radio Burst Catalog (catalog) [pdf].

6. CHIME/FRB Collaboration et al. (2026). The Second CHIME/FRB Catalog of Fast Radio Bursts (canfar data release) [pdf].

7. CHIME/FRB Collaboration, fitburst package (github.com/CHIMEFRB/fitburst), physics engine used to model each simulated burst's dynamic spectrum (Section 3).